

Advanced Spleen Segmentation in 3D CT Scans Using Transformer-Based and Semi-Supervised Learning

Project for HKUST ELEC4840: Artificial Intelligence for Medical Image Analysis (Spring 2024-25)

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Abstract

Accurate spleen segmentation in computed tomography (CT) images is crucial for diagnosing conditions like liver diseases and blood cancers, and for planning medical interventions. However, the 3D U-Net used in Assignment 2 struggles to capture depth features from NIfTI (.nii.gz) files due to its architectural limitations. This study aims to improve spleen segmentation by implementing novel methods, using 3D U-Net of Assignment 2 as a baseline. We adapted SwinUNET[1][2] to process NIfTI files, applied semi-supervised learning with UniMatch [3] on SwinUNET, fine-tuned SAM-Med3D [4], and performed zero-shot inference with MedSAM2[5][6]. All four methods significantly outperformed the baseline 3D U-Net in dice coefficient, Jaccard index, average surface distance (ASD), and 95% Hausdorff Distance (95HD). These advancements demonstrate the potential of transformer-based and advanced learning techniques to enhance 3D medical image segmentation.

1 Introduction

The spleen plays a crucial role in immune surveillance and blood filtration, making its accurate segmentation from thorax-abdomen CT scans vital for clinical tasks such as splenomegaly assessment and treatment planning. CT provides high-resolution volumetric data for detailed organ analysis, thus, reliable spleen segmentation is fundamental for effective clinical decision-making.

Deep learning has transformed automated medical image segmentation. U-Net and its 3D variants [7] are widely used baselines, while SwinUNET[1][2] further improves performance by capturing long-range dependencies using transformer blocks. However, limited annotated 3D data constrains supervised approaches. Semi-supervised frameworks like UniMatch [3] leverage unlabeled data to address this, though their use in 3D segmentation is still emerging. Foundation models such as SAM-Med3D [4] and MedSAM2[5][6] further enable robust fine-tuning and zero-shot inference across diverse tasks.

Our previous 3D U-Net baseline achieved moderate results (Table B.1), but suffered from limited receptive field and insufficient data, leading to suboptimal accuracy and boundary delineation. This report aims to improve 3D spleen segmentation by building on these advanced methods. We compare SwinUNET, adapt UniMatch for semi-supervised 3D segmentation, and evaluate SAM-Med3D fine-tuning and MedSAM2 zero-shot inference, seeking the most effective strategy for accurate and robust spleen segmentation.

2 Methodology

All methodologies are implemented in the Google Colab environment with access to an A100 GPU. Random seeds are set throughout the experiments to ensure reproducibility of the results.

2.1 SwinUNET[1][2]

SwinUNET (Fig.A.2) is a state-of-the-art neural network architecture for medical image segmentation that combines the strengths of convolutional neural networks and transformers. At its core, SwinUNET utilizes the Swin Transformer as its feature extractor, which introduces a novel attention mechanism based on shifted windows. This allows the model to efficiently capture both local and global contextual information within 3D medical images, which is essential for accurately segmenting complex anatomical structures.

Compared to the basic 3D UNet, which relies solely on convolutional layers, SwinUNET offers several important advantages. While 3D UNet is effective at learning local features, it can struggle to capture long-range dependencies within large volumetric data. SwinUNET addresses this limitation by using transformer blocks that model relationships between distant regions in the image. Additionally, the hierarchical design of SwinUNET enables multi-scale feature extraction, allowing the network to recognize both fine details and broader contextual patterns. As a result, SwinUNET tends to achieve higher segmentation accuracy and better generalization on diverse datasets, making it a more powerful and flexible tool for medical image analysis than the traditional 3D UNet.

Data Preprocessing and Augmentation

For all datasets, images and labels were loaded in channel-first format, normalized, cropped to foreground, reoriented, and resampled to a uniform voxel spacing for consistency. During training, random patch cropping around both positive and negative regions was applied to enhance diversity and address class imbalance. Validation and test sets only underwent standard preprocessing, without random cropping, to ensure fair evaluation.

Model Implementation and Training Settings

The SwinUNETR model from MONAI was used for 3D segmentation, configured with a (96, 96, 96) input size, one input channel, and two output channels. The feature size was 12, with dropout rates set to 0.2 for regularization.

Training was performed for up to 200 epochs with validation every two epochs and early stopping (patience 10) based on the best validation metric. DiceCELoss was used with the AdamW optimizer (learning rate 1×10^{-4} , weight decay 0.1). A two-stage learning rate schedule was used: a linear warmup over the first 10 epochs, increasing the rate from 1×10^{-6} to the base value, followed by a ReduceLROnPlateau scheduler that adaptively lowered the rate when validation performance plateaued. This approach supported stable convergence and contributed to the strong performance of the SwinUNETR model on the segmentation task.

2.2 Semi-supervised Learning with UniMatch[3] on SwinUNET[1][2]

To further improve SwinUNET’s performance, we adapted the UniMatch semi-supervised learning framework (Fig.A.5) to 3D spleen segmentation. UniMatch, originally designed for 2D tasks, leverages both labeled and unlabeled data by enforcing consistency between model predictions under different input perturbations, using pseudo-labels to guide learning on unlabeled samples. The overall training objective combines supervised loss on labeled data with an unsupervised consistency loss (Equ.C.1) on unlabeled data, making UniMatch particularly effective in scenarios where labeled examples are scarce.

Our key innovation is extending UniMatch’s dual-stream perturbation and augmentation strategies to volumetric data, and modifying SwinUNET to support feature-level perturbations. This integration enables more robust learning from limited labeled 3D CT scans, enhancing generalization and segmentation accuracy in data-scarce clinical settings.

Dataset

For our experiments, we use all labeled data provided in Assignment 2 as the supervised training and validation set. For validation dataset, the data is preprocessed as the fully supervised SwinUNET approach did. Additionally, we incorporate 100 unlabeled cases randomly sampled from the AbdomenCT-1K dataset to serve as the unsupervised component (Fig.A.1). To ensure training efficiency and balance the number of labeled and unlabeled samples per training epoch, we employ a RandomSampler with replacement on the unlabeled set, ensuring that the number of unlabeled samples matches the size of the labeled dataset in each epoch.

Data Augmentation (Perturbation) and Model Adaptation :

- **Image-level perturbation** For both labeled and unlabeled data, weak perturbation consists solely of random cropping using the MONAI package, introducing moderate spatial variation while preserving anatomical structure for reliable pseudo-labeling. For unlabeled data, strong augmentations (MONAI—random flipping, affine transforms, intensity shifts, Gaussian noise, and smoothing) introduce substantial spatial and intensity variation. This encourages the model to maintain prediction consistency under significant input changes, enhancing robustness and generalization.
- **Feature-level perturbation:** Following the approach described in the original UniMatch paper—which applies Dropout2D with a probability of 0.5—we extend this idea to the 3D context by employing Dropout3D with a probability of 0.5. Specifically, we apply Dropout3D to the output feature maps from both encoder 1 and encoder 3 within the SwinUNET architecture. This introduces stochastic noise directly at the feature level, encouraging the network to learn more robust and invariant representations.
- **Model adaptation:** To adapt the SwinUNETR architecture for the UniMatch framework, we introduce a modified forward method (Fig. A.3) that enables feature-level perturbation. Specifically, the model extracts feature maps from both the first and the deepest encoder layers. When feature perturbation is required, Dropout3D with a probability of 0.5 is applied to these feature maps before they are passed to the decoder. The model then computes outputs using both the original and perturbed features, allowing the consistency loss to be enforced at the feature level.

Training setting

For training, we use the same model configuration as in the fully supervised SwinUNETR approach. The model is first pretrained on the labeled dataset using the fully supervised setting, and the checkpoint with the best validation Dice score is loaded before semi-supervised training begins. We employ a **CosineAnnealingLR** scheduler with $T_{\max} = 50$ and use DiceCELoss for both supervised and unsupervised losses. For the unsupervised component, only pseudo labels with at least 95% confidence from weakly-perturbed unlabeled images are used. After testing with various unsupervised loss weights, the unified loss function is defined as in equation (Equ.C.2.)

Performance is evaluated on the validation set during training. Additional settings include early stopping with a patience of 5, experiment logging, a batch size of 4, a maximum of 50 epochs, initial learning rate 1×10^{-4} , learning rate decay ($\lambda = 0.9$), and a crop size of $160 \times 160 \times 160$ for sliding window inference during validation.

2.3 SAM-Med3D fine-tuning[4]

SAM-Med3D (Fig.A.6) is a foundation model designed for medical image segmentation, built upon the Segment Anything Model (SAM) architecture and pre-trained on large-scale 3D medical imaging datasets. As a foundation model, SAM-Med3D leverages

extensive pretraining to learn generalizable visual representations, making it highly effective for downstream tasks such as organ and lesion segmentation across diverse modalities and anatomies. The primary strength of SAM-Med3D lies in its ability to adapt quickly to new medical segmentation tasks with limited annotated data, thanks to its robust and transferable feature representations. By finetuning SAM-Med3D on our specific dataset, we can benefit from improved accuracy, faster convergence, and enhanced generalization compared to training a model from scratch.

Data Preprocessing

Before fine-tuning, all images and labels are standardized for compatibility with SAM-Med3D. Preprocessing involves: (1) resampling NIfTI images and labels to (1.5, 1.5, 1.5) mm voxel spacing using TorchIO for spatial consistency; (2) generating binary masks for each target class by assigning 1 to relevant voxels and 0 elsewhere; and (3) excluding cases with segmented object volumes below 10 mm³ to remove small or noisy regions.

Finetuning Procedure

For finetuning, we follow the official training code provided by the SAM-Med3D GitHub repository. To enhance the robustness of the training process, we add a validation loop and implement early stopping with a patience of 5 epochs. During finetuning, the `image_encoder` and `prompt_encoder` components of the model are kept frozen, and only the mask decoder is updated. The model is trained for up to 30 epochs with a batch size of 8, while all other hyperparameters are left at their default settings as specified in the official implementation. This setup helps prevent overfitting and ensures efficient use of computational resources during the finetuning process.

2.4 MedSAM2 Zero-shot Inference[5][6]

MedSAM2 is a promptable foundation model designed for 3D medical image segmentation, addressing depth (Z-axis) feature challenges inherent in volumetric CT data. Its Network Architecture(Fig.A.7) comprises four components: (1) a Hierarchical Vision Transformer (Hiera) image encoder that processes 2D slices to extract multi-scale spatial features, avoiding the computational burden of 3D convolutions while preserving anatomical details; (2) a memory attention module that maintains a streaming memory bank of prior slice features, enabling spatial continuity across the Z-axis via self- and cross-attention mechanisms; (3) a prompt encoder that converts user inputs (e.g., bounding boxes on a middle slice) into embeddings to guide segmentation; and (4) a mask decoder that fuses multi-scale features and memory-conditioned context to generate high-resolution 3D masks. For spleen segmentation, MedSAM2 employs zero-shot inference: a bounding box prompt on the middle slice initiates bidirectional propagation, leveraging memory-driven spatial coherence to ensure consistent segmentation across adjacent slices. Compared to 3D U-Net, which struggles with depth-wise inconsistencies due to fixed convolutional kernels and limited receptive fields, MedSAM2 resolves these issues through transformer-based feature extraction and explicit modeling of slice-to-slice dependencies. The memory attention module propagates contextual cues (e.g., spleen shape and position) across slices, minimizing fragmented outputs.

To perform spleen segmentation in 3D CT scans, we implemented MedSAM2 using the provided checkpoint (MedSAM2_latest.pt) and configuration file (sam2.1_hiera_t512.yaml), leverages a Hierarchical Vision Transformer (Hiera) image encoder. Zero-shot inference further eliminates the need for task-specific retraining, enabling generalization to unseen anatomies. This methodology combines computational efficiency with robust 3D consistency, offering a scalable solution for clinical spleen segmentation.

3 Result analysis

3.1 The SwinUNET[1][2] and The Semi-supervised Learning with UniMatch [3] vs. Baseline of 3D UNet

According to the Table B.3, The SwinUNET achieved a Dice Coefficient of 0.8045, representing a modest improvement of 1.71% over the 3D UNet’s score of 0.7874, indicating better overlap between predicted and ground truth segmentations. Similarly, the Jaccard Index improved from 0.6580 (3D UNet) to 0.6787 (SwinUNET), a relative increase of 2.07%, further supporting the enhanced segmentation accuracy of the SwinUNET.

However, the results for boundary-based metrics present a more nuanced picture. The Average Surface Distance (ASD) for SwinUNET slightly worsened (10.9014) compared to the 3D UNet (10.4824), with an increase of 0.4190, suggesting that although SwinUNET produces more accurate volumetric overlaps, its boundary predictions may be less precise or more irregular. This trend is also reflected in the 95% Hausdorff Distance, where SwinUNET recorded a higher value (71.7465) than the 3D UNet (52.5198), indicating larger outlier surface distances in some cases.

The training and validation curves of Loss and Dice in Fig.A.8 provide further insight into SwinUNET’s performance. Both loss and Dice scores improve steadily over 70 epochs, with the validation loss closely tracking the training loss. The validation Dice score stabilizes above 0.7, indicating robust segmentation performance and effective learning. The small gap between training and validation metrics suggests that SwinUNET generalizes well and does not suffer from severe overfitting.

In summary, while SwinUNET demonstrates improvements in overall segmentation accuracy as measured by Dice and Jaccard scores, it does not consistently outperform the 3D UNet in terms of boundary delineation. These results suggest that transformer-based architectures like SwinUNET can offer better region-level segmentation but may require further optimization to achieve superior boundary precision in medical image segmentation tasks.

We further evaluated SwinUNET using the UniMatch semi-supervised framework under different training configurations, as shown in Fig.B.2. Lowering the weight of the unsupervised loss led to a notable improvement in segmentation performance,

with the Dice score increasing from 0.7995 to 0.8241 without CutMix. CutMix is a data augmentation technique that enhances model generalization by combining two training images. Introducing CutMix augmentation with lower unsupervised loss weights also improved performance compared to higher loss weighting, achieving a Dice score of 0.8197, though it did not surpass the best setting without CutMix. These results indicate that careful tuning of the unsupervised loss weight is crucial for optimal semi-supervised learning, and that while CutMix can be beneficial, its impact is sensitive to other hyperparameters. Overall, the findings highlight the importance of balancing supervised and unsupervised objectives when applying semi-supervised methods to transformer-based segmentation models.

Yet, the improvement from semi-supervised learning is often limited by the quality of pseudo labels generated for the unlabeled data, as illustrated in Fig.A.4a and A.4b. Noisy or inaccurate pseudo labels can misguide the model during training, reducing the effectiveness of the unsupervised component. Overall, our findings underscore the need to balance supervised and unsupervised objectives and ensure pseudo label reliability when applying semi-supervised methods.

Comparison of SwinUNET and Semi-supervised UniMatch-SwinUNET

Table B.3 presents a performance comparison between the baseline 3D U-Net, SwinUNET, and SwinUNET trained with UniMatch semi-supervised learning. Notably, applying UniMatch further boosts performance: UniMatch-SwinUNET achieves the highest Dice (0.8241), Jaccard (0.7061), and shows substantial improvements in ASD and 95% HD compared to the baseline and SwinUNET. Specifically, UniMatch-SwinUNET surpasses fully supervised SwinUNET by 2.4% in Dice and 4.0% in Jaccard, while also significantly reducing ASD and 95% HD. Semi-supervised learning with UniMatch can improve these distance-based metrics as it leverages additional unlabeled data to provide more diverse and informative training examples. This helps the model learn more accurate and robust boundary representations. These results demonstrate that incorporating semi-supervised learning with UniMatch leads to better segmentation performance than using fully supervised SwinUNET alone.

3.2 SAM-Med3D fine-tuning[4] and MedSAM2 Zero-shot Inference[5][6]

Fine-tuning SAM-Med3D leads to substantial segmentation improvements compared to the 3D U-Net baseline. According to Table B.4, for the Dice coefficient, SAM-Med3D achieves 0.9398 before fine-tuning and 0.9399 after, both notably higher than the 3D U-Net’s 0.7874, reflecting a relative improvement of over 15%. Similarly, the Jaccard Index rises from 0.6580 (3D U-Net) to 0.8869 (SAM-Med3D before fine-tuning) and 0.8872 (after), yielding a gain of nearly 23%. Surface distance metrics also show dramatic enhancements: Average Surface Distance (ASD) drops from 10.48 to around 0.65, and 95% Hausdorff Distance (95HD) plummets from 52.52 to approximately 1.76. These results highlight that SAM-Med3D, even before fine-tuning, significantly outperforms the baseline, and fine-tuning yields marginal but consistent further gains across all metrics.

MedSAM2 also demonstrates substantial improvements over the 3D U-Net baseline across all evaluation metrics (Table B.5). The Dice coefficient increases from 0.7874 to 0.9579 (a 17.05% improvement), and the Jaccard Index rises from 0.6580 to 0.9193 (a 26.13% improvement). For boundary accuracy, the Average Surface Distance (ASD) drops significantly from 10.48 to 0.31, and the 95% Hausdorff Distance (95HD) decreases from 52.52 to 1.10. These results highlight that MedSAM2 delivers much more accurate and reliable segmentations than the baseline model.

The large performance gains of foundation models like SAM-Med3D and MedSAM2 are mainly due to their extensive pretraining on diverse medical datasets. This enables them to learn robust, generalizable features and adapt quickly to new tasks, resulting in much higher segmentation accuracy compared to models trained from scratch.

3.3 Overall Performance Analysis across All Methodologies

Quantitative Metrics Comparison

As shown in Table B.6, SwinUNET provides only minor improvements over the baseline 3D U-Net, with slight increases in Dice and Jaccard but worse ASD and 95% HD. Incorporating semi-supervised learning with UniMatch yields moderate gains in Dice and Jaccard, and substantial improvements in ASD and 95% HD. Foundation models, such as SAM-Med3D and MedSAM2, achieve the largest performance boosts across all metrics, with MedSAM2 (zero-shot) performing best overall. The results highlight that while semi-supervised approaches offer some benefits, foundation models deliver the most significant advances in segmentation accuracy and robustness.

Segmentation Visualization Comparison

Visualization of the predicted segmentation masks in Fig.A.12 reveals that all proposed methodologies outperform the baseline 3D U-Net, producing more accurate segmentations across various samples. SwinUNET shows improved volume-based metrics compared to the 3D U-Net, indicating better overall coverage; however, its distance-based metrics are less favorable, as indicated by less precise organ boundaries in certain visualizations. UniMatch-SwinUNET further enhances SwinUNET’s performance in both volume- and distance-based metrics, with these improvements particularly evident in sample spleen 63, where boundary accuracy is notably higher (Fig. A.9). Among all methods, SAM-Med3D and MedSAM2 produce the most accurate segmentations, with the best boundary delineation and overlap with ground truth. Notably, MedSAM2 achieves the fewest false positive overlaps and the most precise boundaries. In summary, MedSAM2 and SAM-Med3D provide the highest visual and quantitative segmentation performance, followed by UniMatch-SwinUNET, SwinUNET, and finally the baseline 3D U-Net.

4 Conclusion

In this work, we explored the use of SwinUNET for spleen segmentation on 3D CT data and introduced an innovation by adapting the model to the UniMatch framework for semi-supervised learning. This allowed us to leverage unlabeled data through pseudo-labeling and capture long-range dependencies across the image volume, resulting in a modest performance boost despite the challenges posed by variable pseudo-label quality. Additionally, we fine-tuned SAM-Med3D and performed zero-shot inference with MedSAM2, both of which achieved significant improvements in segmentation performance, likely due to their strong pretrained representations and robust generalization. For future enhancement, potential directions include exploring SAM adapters, developing multi-model ensembles to exploit complementary model strengths, and investigating alternative semi-supervised learning methodologies.

References

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- [3] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, “Revisiting weak-to-strong consistency in semi-supervised semantic segmentation,” in *CVPR*, 2023.
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You may include other additional sections here.

A Appendix: Figure

All the figures have been used:

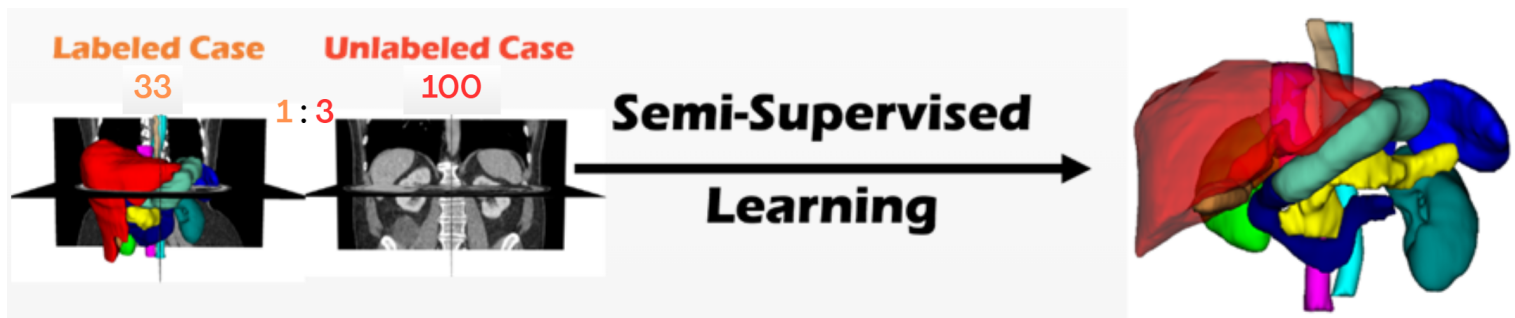


Figure A.1: The labeled SpleenDataset and unlabeled AbdomenCT-1K dataset

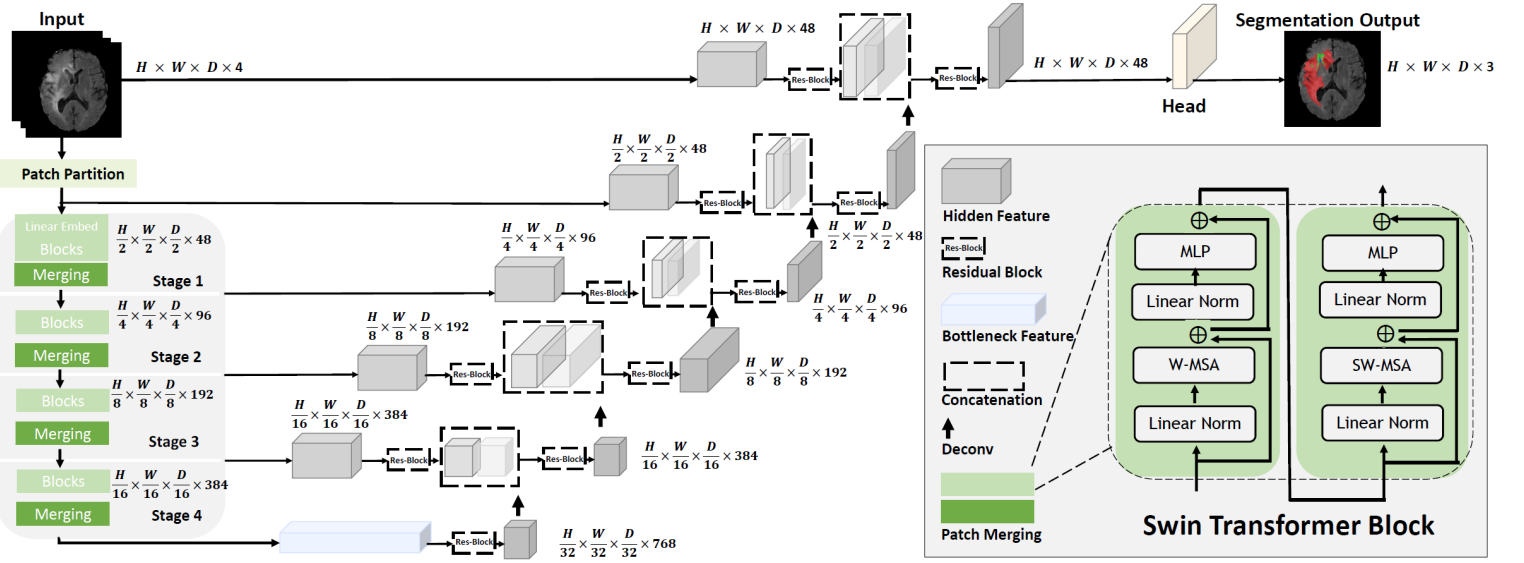


Figure A.2: **Overview of the Swin UNETR architecture**[2]. The input to our model is 3D multi-modal MRI images with 4 channels. The Swin UNETR creates non-overlapping patches of the input data and uses a patch partition layer to create windows with a desired size for computing the self-attention. The encoded feature representations in the Swin transformer are fed to a CNN-decoder via skip connection at multiple resolutions.

```

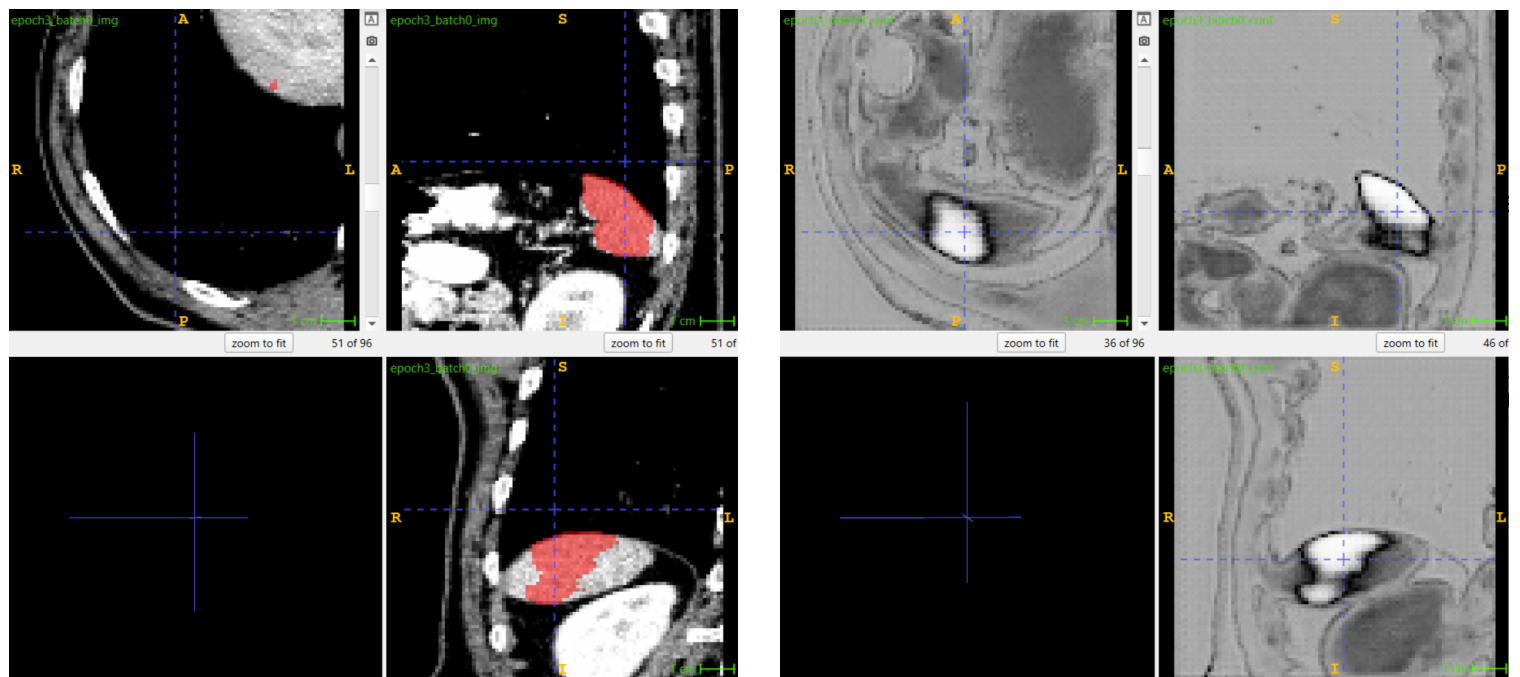
if need_fp:
    # Apply perturbation if provided
    enc0_fp = perturb_fn(enc0) if perturb_fn else nn.Dropout3d(0.5)(enc0)
    enc3_fp = perturb_fn(enc3) if perturb_fn else nn.Dropout3d(0.5)(enc3)

    # Get logits for both original and perturbed features
    logits = self.decode_and_classify(enc0, enc3, hidden_states_out)
    logits_fp = self.decode_and_classify(enc0_fp, enc3_fp, hidden_states_out)
    return logits, logits_fp

# Only original features
logits = self.decode_and_classify(enc0, enc3, hidden_states_out)
return logits

```

Figure A.3: **SwinUNET Model Adaption for the UniMatch Supervision.**



(a) Visualization of the Pseudo Label Quality on Unlabeled Dataset for Training

(b) Visualization of the Pseudo Label Confidence Quality on Unlabeled Dataset for Training

Figure A.4: Visualization of Pseudo Label and Confidence Quality on Unlabeled Dataset for Training

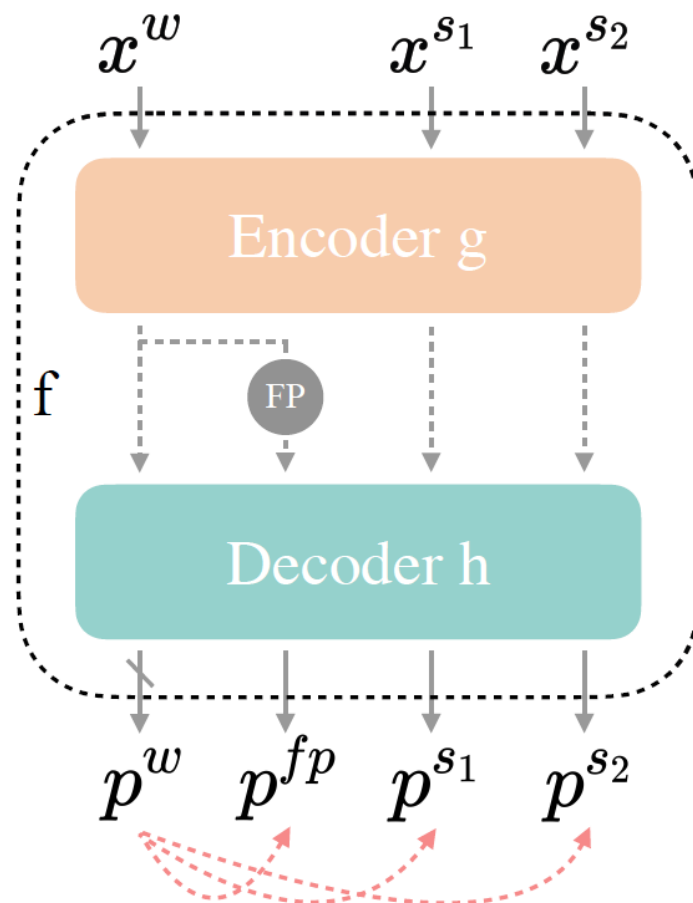


Figure A.5: **UniMatch** [3]: The unified dual-stream perturbations method. The FP denotes feature perturbation, and the dashed curves represent supervision.

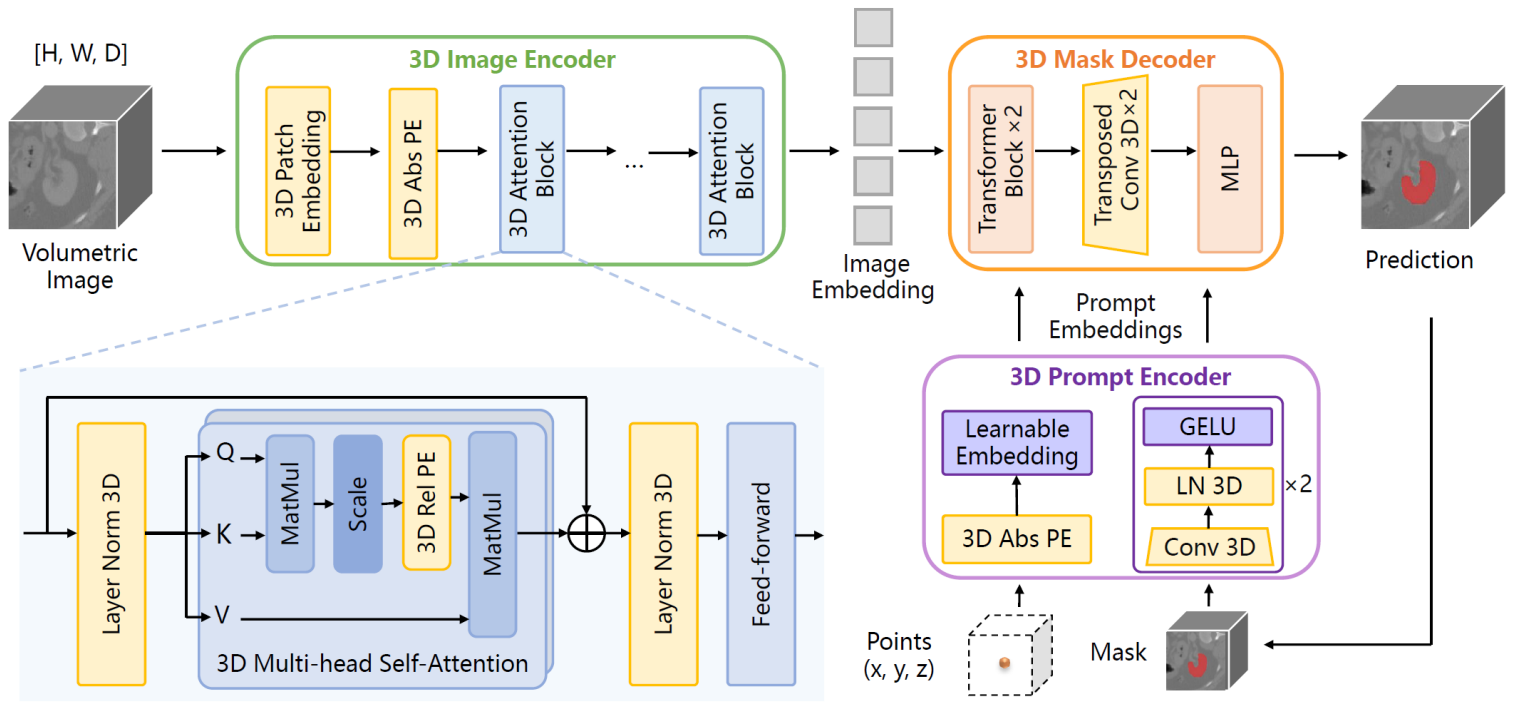


Figure A.6: **The Fully 3D Architecture of the SAM-Med3D[4]**, encompassing a 3D image en-coder, 3D prompt encoder, and 3D mask decoder. 3D positional encoding (PE) and 3D layers like convolution and layer normalization are employed to construct it.

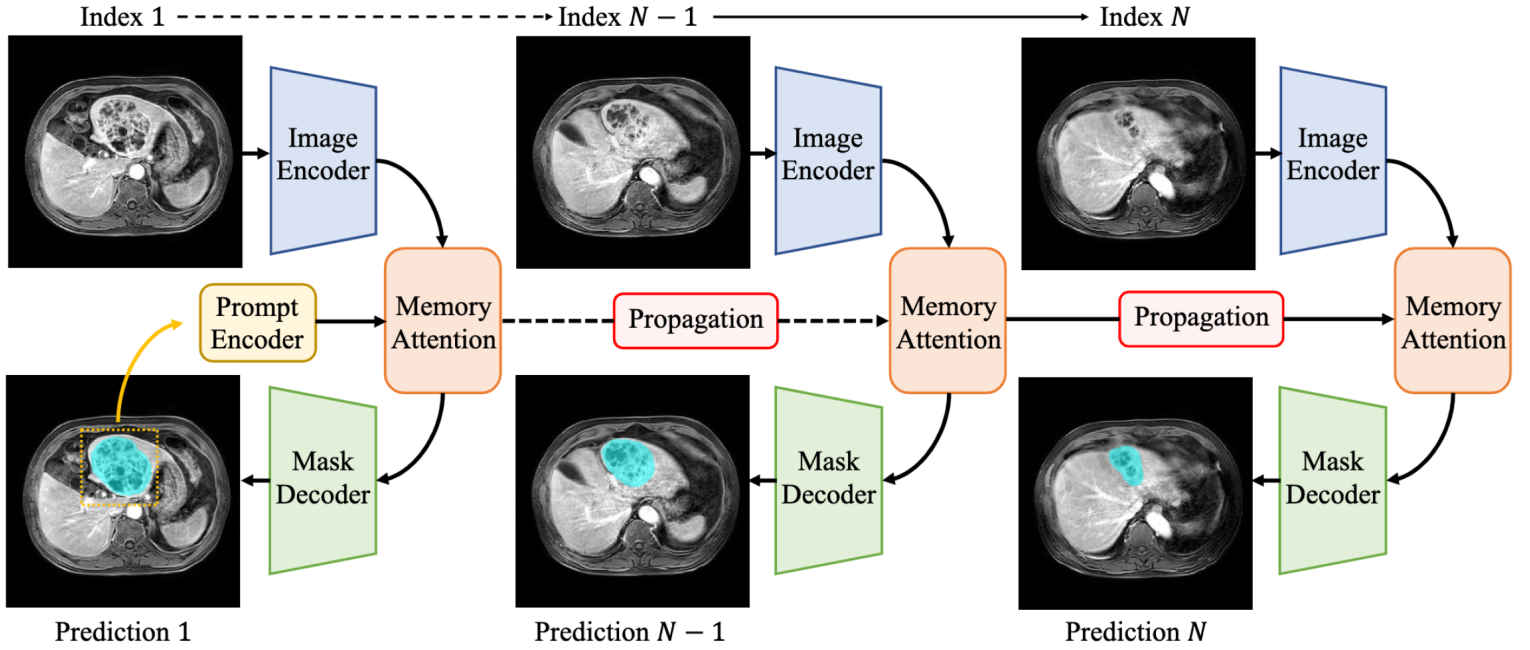
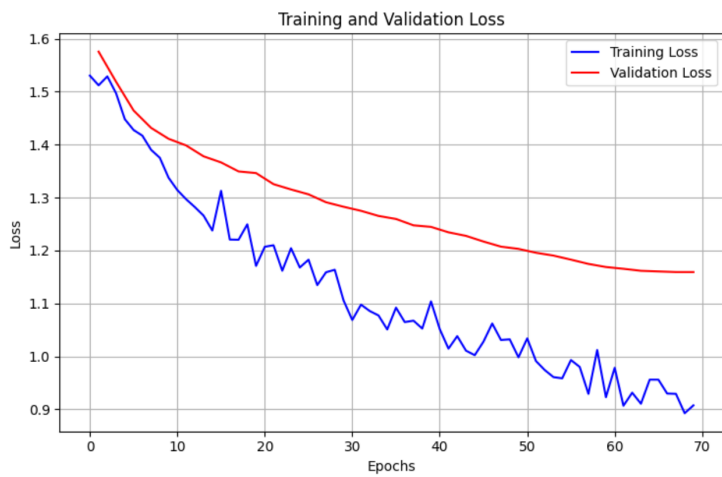
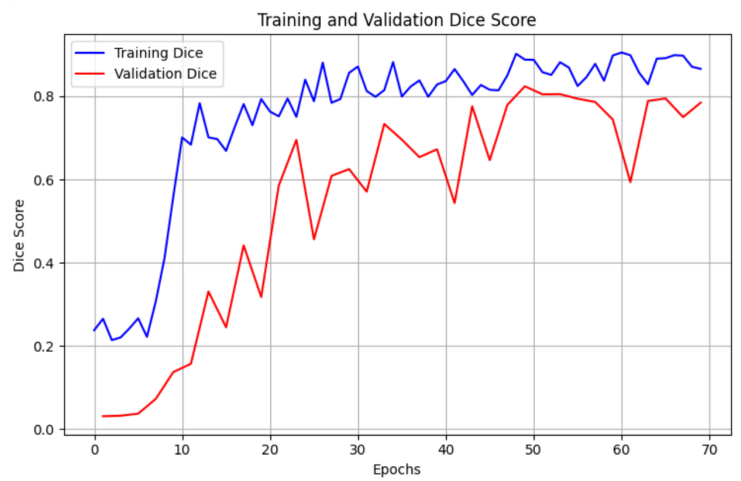


Figure A.7: **The Network Architecture for MedSAM2 Development[5]**: MedSAM2 is a promptable segmentation network with an image encoder, a prompt encoder, a memory attention module, and a mask decoder. The image encoder extracts multiscale features from each frame or 2D slice. The memory attention module conditions the current frame features on past frames' features and predictions using streaming memory. The mask decoder generates accurate segmentation masks based on bounding box prompts and memory-conditioned features. This architecture enables MedSAM2 to effectively segment both 3D medical images and videos by exploiting spatial continuity across slices and frames.



(a) Train and Validation Loss of The SwinUNET



(b) Train and Validation Dice of SAM-Med3D After Fine-Tuning

Figure A.8: Train & Validation Loss and Dice of The SwinUNET

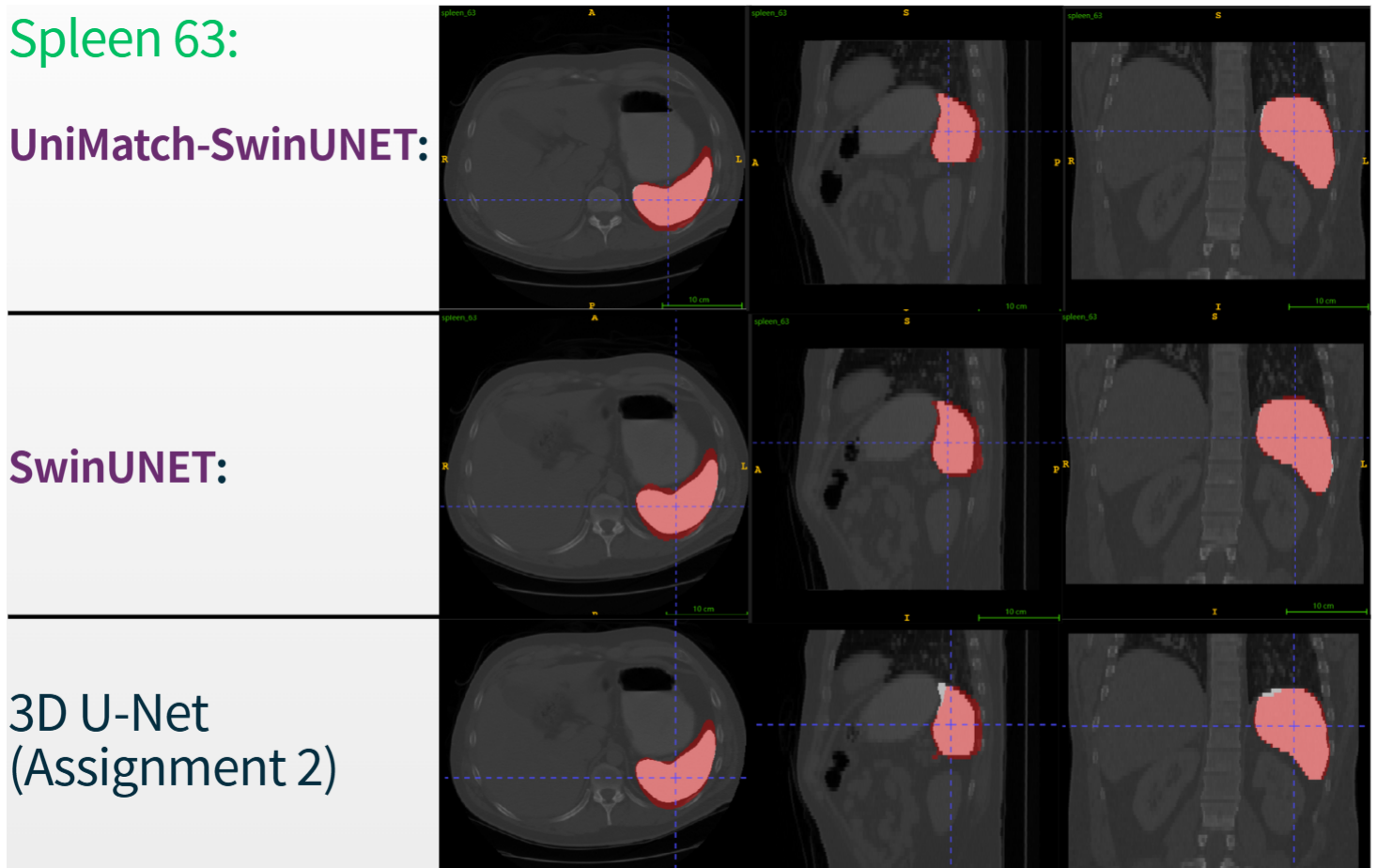


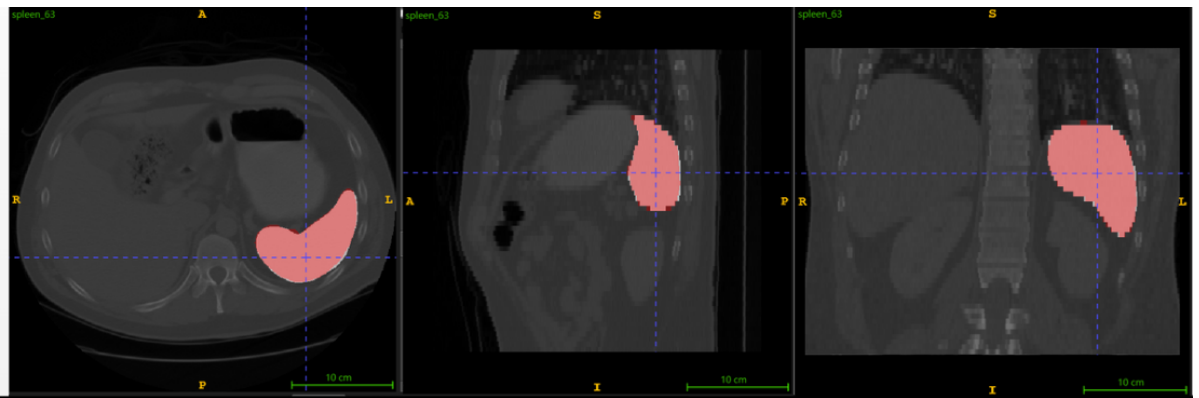
Figure A.9: ITK-SNAP Visualisation Comparison of 3D U-Net, SwinUNET, UniMatch-SwinUNET on the Spleen 63 of Test Sets

Red Region: Segmentation Prediction

White Region: Ground Truth

Spleen 63:

SAM-Med3D



3D U-Net
(Assignment 2)

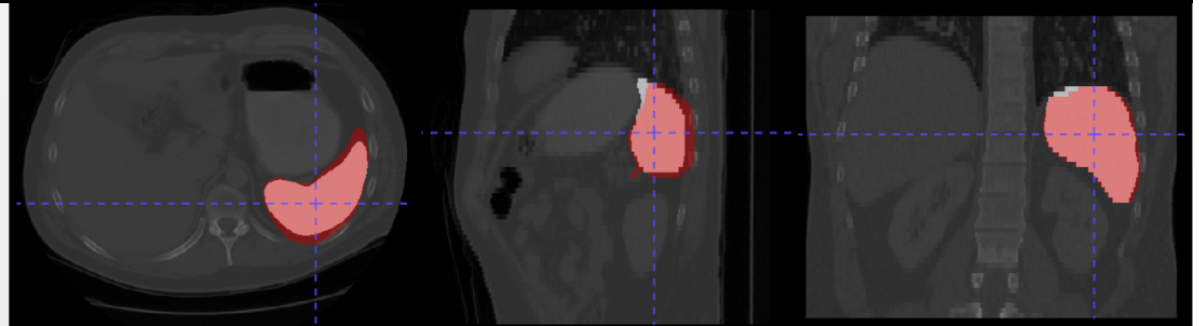


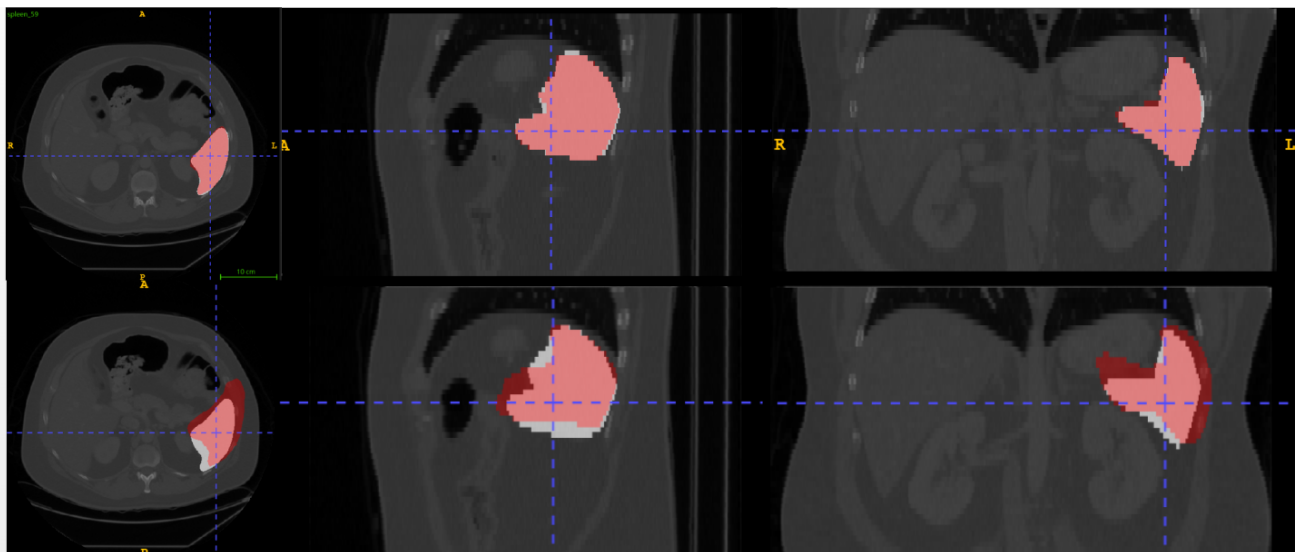
Figure A.10: ITK-SNAP Visualisation Comparison between 3D U-Net and SAM-Med3D on the Spleen 63 of Test Sets

Red Region: Segmentation Prediction

White Region: Ground Truth

Spleen 59:

MedSAM2



3D U-Net
(Assignment 2)

Figure A.11: ITK-SNAP Visualisation Comparison between 3D U-Net and MedSAM2 on the Spleen 59 of Test Sets

Red Region: Segmentation Prediction

White Region: Ground Truth

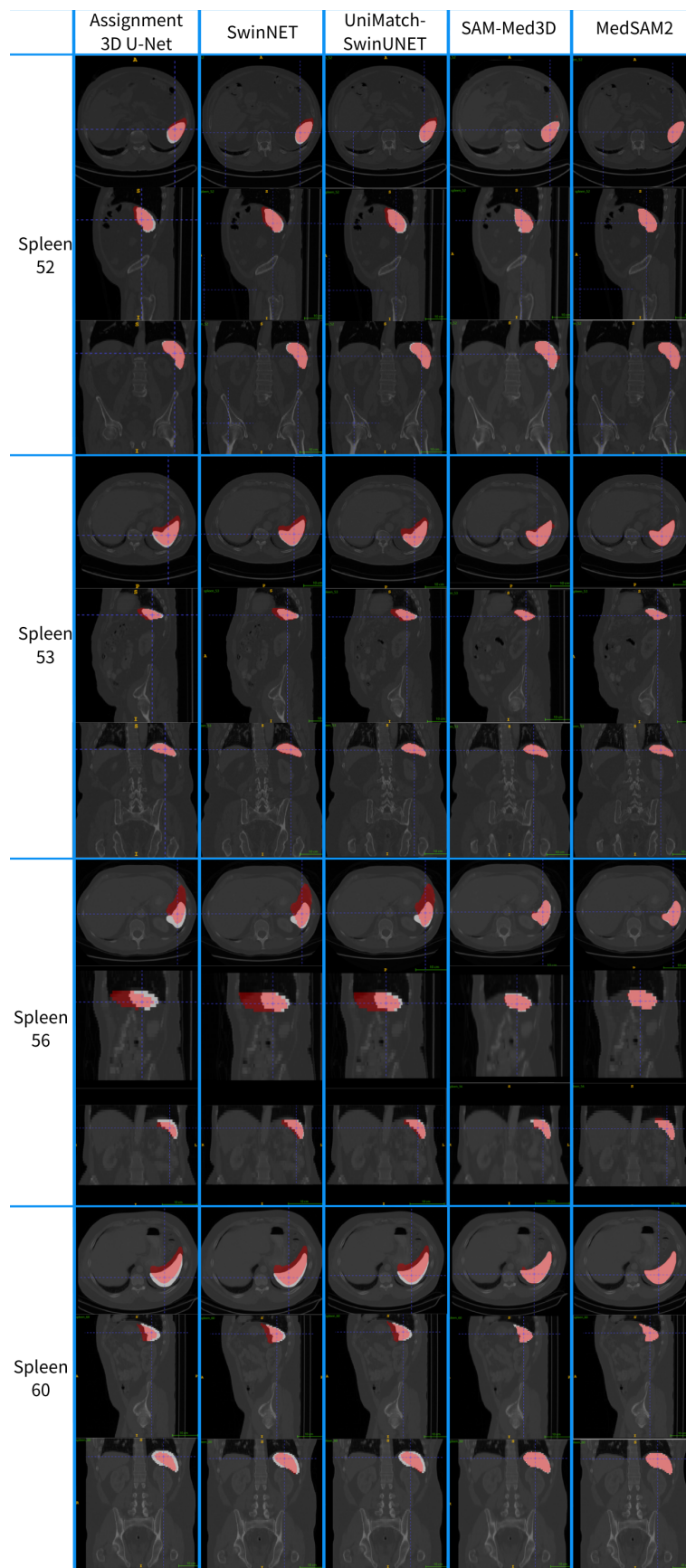


Figure A.12: **ITK-SNAP Visualisation Comparison All Model on the Spleen 52, 53, 56 and 60 of Test Sets**
Red Region: Segmentation Prediction
White Region: Ground Truth

B Appendix: Table

All the tables have been used:

Table B.1: **Baseline Performance Based on the 3D U-Net of Assignment 2**

Segmentation performance is evaluated using Dice Similarity Coefficient (DSC) for volumetric overlap, Jaccard Index for mask precision, Average Surface Distance (ASD) for boundary alignment, and 95% Percentile Hausdorff Distance (95HD) for worst-case errors.

	Dice	Jaccard	ASD	95% Hausdorff Dist
3D U-Net’s Evaluation Metrics Result	0.7874	0.6580	10.4824	52.5198

Table B.2: **Different Settings and Performance Comparison on UniMatch-SwinUNET (weights & CutMix)**

Setting	Dice	@Epoch stopped
Higher weights for unsupervised loss w/o CutMix	0.7995	12
Lower weights for unsupervised loss w/o CutMix	0.8241	15
Lower weights for unsupervised loss + CutMix	0.8197	6

Table B.3: **Performance Comparison of 3D U-Net, SwinUNET and Semi-supervised Learning with UniMatch on SwinUNET**

Model	Dice (↑)	Jaccard (↑)	ASD (↓)	95% HD (↓)
Assignment 2 – 3D U-Net	0.7874	0.6580	10.4824	52.5198
SwinUNET	0.8045 (+2.1717%)	0.6787 (+3.1459%)	10.9014 (+0.4190)	71.7465 (+19.2267)
UniMatch-SwinUNET	0.8241 (+4.661%)	0.7061 (+7.310%)	4.7443 (-5.7381)	31.5721 (-20.9477)

Table B.4: **Performance Comparison of 3D U-Net and SAM-Med3D Models**

Method	Dice (↑)	Jaccard (↑)	ASD (↓)	95% HD (↓)
3D U-Net of the Assignment 2	0.7874	0.6580	10.4824	52.5198
SAM-Med3D Before Fine-Tuning	0.9398 (+15.244%)	0.8869 (+22.89%)	0.6438 (-9.8386)	1.8059 (-50.7139)
SAM-Med3D After Fine-Tuning	0.9399 (+15.25%)	0.8872 (+22.92%)	0.6611 (-9.8213)	1.7593 (-50.7605)

Table B.5: **Performance Comparison between Assignment 2 and MedSAM2[5][6]**

Method	Dice (↑)	Jaccard (↑)	ASD (↓)	95% HD (↓)
Assignment 2	0.7874	0.6580	10.4824	52.5198
MedSAM2	0.9579 (+17.05%)	0.9193 (+26.13%)	0.3079 (-10.1745)	1.1036 (-51.4162)

C Appendix: Equation

$$\mathcal{L}_u = \frac{1}{B_u} \sum \mathbb{1}(\max(p^w) \geq \tau) \cdot \left(\lambda H(p^w, p^{fp}) + \frac{\mu}{2} (H(p^w, p^{s1}) + H(p^w, p^{s2})) \right)^1 \quad (\text{C.1})$$

1

- $\mathcal{L}_u$: Unsupervised loss function, used to train the model on unlabeled data by enforcing consistency across perturbations.
- B_u : Batch size of unlabeled data, used to normalize the loss by averaging over the batch.
- $\sum$: Summation over all samples in the batch of unlabeled data (index implied).
- $\mathbb{1}$: Indicator function, equals 1 if the condition $\max(p^w) \geq \tau$ is true, 0 otherwise, filtering uncertain predictions.
- p^w : Predicted probability distribution from the weakly augmented input (e.g., with minor transformations like cropping).
- $\max(p^w)$: Maximum probability in p^w , indicating the model’s confidence in its prediction for the weakly augmented input.
- τ : Confidence threshold (e.g., 0.95 for Pascal dataset), ensures only confident predictions contribute to the loss.
- λ : Weight for the feature-level consistency loss term (set to 0.5 in the paper).

Table B.6: **Performance Comparison of All Methods with Improvements Relative to Assignment 2 – 3D U-Net**
 Segmentation performance is evaluated using Dice Similarity Coefficient (DSC) for volumetric overlap, Jaccard Index for mask precision, Average Surface Distance (ASD) for boundary alignment, and 95% Percentile Hausdorff Distance (95HD) for worst-case errors.

Model	Dice (↑)	Jaccard (↑)	ASD (↓)	95% HD (↓)
Assignment 2 – 3D U-Net	0.7874	0.6580	10.4824	52.5198
SwinUNET	0.8045 (+2.1717%)	0.6787 (+3.1459%)	10.9014 (+0.419)	71.7465 (+19.2267)
Semi-supervised learning with UniMatch – SwinUNET	0.8241 (+4.661%)	0.7061 (+7.310%)	4.7443 (-5.7381)	31.5721 (-20.9477)
SAM-Med3D Before Fine-Tuning	0.9398 (+19.355%)	0.8869 (+34.787%)	0.6438 (-9.8386)	1.8059 (-50.7139)
SAM-Med3D After Fine-Tuning	0.9399 (+19.368%)	0.8872 (+34.833%)	0.6611 (-9.8213)	1.7593 (-50.7605)
Zero-shot inference with MedSAM2	0.9579 (+21.652%)	0.9193 (+39.711%)	0.3079 (-10.1745)	1.1036 (-51.4162)

$$\text{loss} = \frac{\text{loss}_x + 0.01 \cdot \text{loss}_{u_s1} + 0.01 \cdot \text{loss}_{u_s2} + 0.05 \cdot \text{loss}_{u_w_fp}}{2.0} \tag{C.2}$$

D Appendix: Supplemental Information

Segmentation performance is evaluated using Dice Similarity Coefficient (DSC) for volumetric overlap, Jaccard Index for mask precision, Average Surface Distance (ASD) for boundary alignment, and 95% Percentile Hausdorff Distance (95HD) for worst-case errors.

- H: Cross-entropy loss, measures the difference between two probability distributions.
- p^{fp} : Predicted probability distribution from the feature-perturbed input (e.g., after channel dropout).
- μ : Weight for the weak-to-strong consistency loss term (set to 0.5 in the paper).
- p^{s1} : Predicted probability distribution from the first strongly augmented input (e.g., with heavy transformations like color jitter).
- p^{s2} : Predicted probability distribution from the second strongly augmented input, providing an additional strong view for consistency.